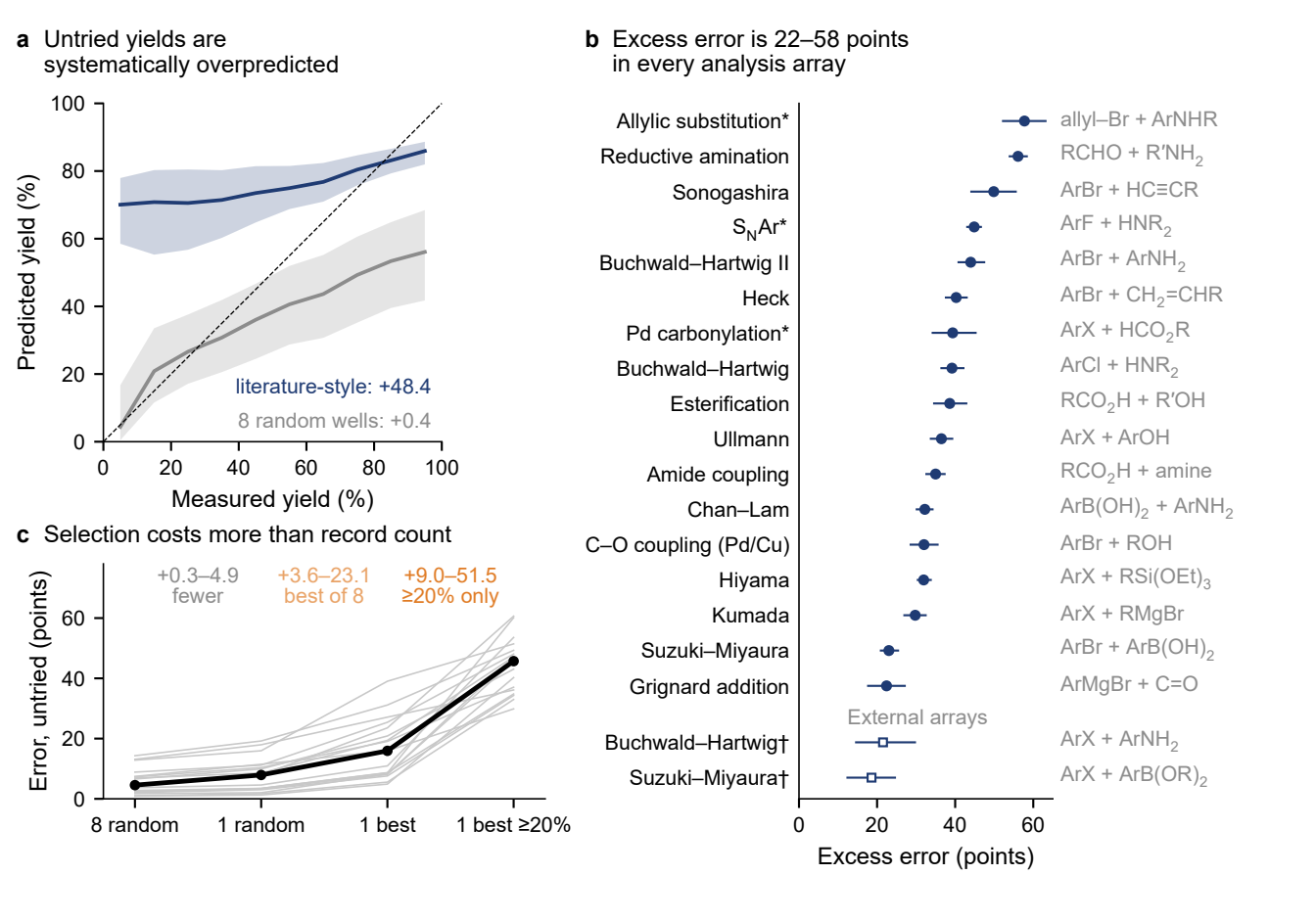
a Untried yields are systematically overpredicted
Predicted yield (%)
Measured yield (%)
literature-style: +48.4
8 random wells: +0.4
b Excess error is 22–58 points in every analysis array
Allylic substitution*
Reductive amination
Sonogashira
S_NAr*
Buchwald–Hartwig II
Heck
Pd carbonylation*
Buchwald–Hartwig
Esterification
Ullmann
Amide coupling
Chan–Lam
C–O coupling (Pd/Cu)
Hiyama
Kumada
Suzuki–Miyaura
Grignard addition
External arrays
Buchwald–Hartwig†
Suzuki–Miyaura†
allyl–Br + ArNHR
RCHO + R′NH2
ArBr + HC≡CR
ArF + HNR2
ArBr + ArNH2
ArBr + CH2=CHR
ArX + HCO2R
ArCl + HNR2
RCO2H + R′OH
ArX + ArOH
RCO2H + amine
ArB(OH)2 + ArNH2
ArBr + ROH
ArX + RSi(OEt)3
ArX + RMgBr
ArBr + ArB(OH)2
ArMgBr + C=O
ArX + ArNH2
ArX + ArB(OR)2
Excess error (points)
c Selection costs more than record count
+0.3–4.9 fewer
+3.6–23.1 best of 8
+9.0–51.5 ≥20% only
Error, untried (points)
8 random
1 random
1 best
1 best ≥20%